3/31

COMPACTNESS

Def: Let X be a top. space. An "open cover" of X is a collection of subsets whose union is equal to X.

X is said to be "compact" if every open cover of X has a finite sub-cover [ie, a finite collection whose union equals X].

Eg: 1. Any finite topological space, and any finite topology on a space.

2. $X = \{0\} \cup \{1/n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ is compact.

Proof: Given any open cover $\{U_\alpha\}_{\alpha \in \Lambda}$ of X,

$\exists\, U_\alpha$ containing 0.

U_α contains $X \cap (-\varepsilon, \varepsilon)$ for some $\varepsilon > 0$.

Pick $N \in \mathbb{N}$ s.t. $1/N < \varepsilon$. Then $\forall\, n \geq N$,

$1/n \in X \cap (-\varepsilon, \varepsilon) \subseteq U_\alpha$.

Now select $U_{\alpha_1}, \ldots, U_{\alpha_{N-1}}$ containing $1, 1/2, \ldots, 1/N-1$ resp.

Then $X = U_\alpha \cup U_{\alpha_1} \cup U_{\alpha_2} \cup \ldots U_{\alpha_{N-1}}$.

Non-Eg: $(0,1]$ is not compact; the open cover $\mathcal{A} = \{(1/n, 1] : n \in \mathbb{N}\}$ has no finite sub-cover.

Proof: given any finite cover $(1/n_1, 1], (1/n_2, 1], \ldots (1/n_k, 1]$,

the union of these sets is $(1/n, 1]$ where $1/n = \min\{1/n_1, \ldots, 1/n_k\}$.

$(1/n, 1] \neq (0,1]$, so $\Rightarrow\Leftarrow$.

Def: Let $Y \subseteq X$. A collection of subsets of X is a "cover of Y" if their union contains Y.

Lemma: Let $Y \subseteq X$. Then Y is compact $\Leftrightarrow$ $\forall$ covering of Y by open sets of X, $\exists$ a finite subcover of Y.

Proof: $\Rightarrow$: Say Y is compact.

Then $\forall$ collection $\mathcal{A} = \{U_\alpha\}_{\alpha \in \Lambda}$ of open sets in X covering Y,

$\{U_\alpha \cap Y\}_{\alpha \in \Lambda}$ is a cover of Y by open sets of Y. This has a finite subcover

$\{U_{\alpha_1} \cap Y, U_{\alpha_2} \cap Y, \ldots, U_{\alpha_n} \cap Y\}$.

$\Rightarrow \bigcup_{i=1}^{n} U_{\alpha_i} \supseteq Y$ is an open sub-cover of Y.

$\Leftarrow$: Exercise.

Theorem: Every closed subspace of a compact space is compact.

Proof: Let $Y \subseteq X$ be closed. Let $\{U_\alpha\}_{\alpha \in \Lambda}$ be a collection of open sets of X that cover Y.

Then let $\mathcal{B} = \{U_\alpha\}_{\alpha \in \Lambda} \cup \{X \setminus Y\}$.

$\mathcal{B}$ is an open cover of X, and has a finite sub-cover. But this is also a finite sub-cover of Y, after discarding $X \setminus Y$ if needed.

Theorem: Every compact subspace of a Hausdorff space is closed.

Proof: Let X Hausdorff space, $Y \subseteq X$ be compact. WTS: Y is closed, ie, $X \setminus Y$ is open.

Let $x_0 \in X \setminus Y$. We show $\exists\, U \ni x_0$ open s.t. $U \subseteq X \setminus Y$.

$\because X$ is Hausdorff, $\forall\, y \in Y$, $\exists\, V_y \ni y$ and $U_y \ni x_0$ (open sets) s.t. $V_y \cap U_y = \emptyset$.

Note: $Y \subseteq \bigcup_{y \in Y} V_y$, so $\exists$ a finite subcover $\{V_{y_1}, \ldots, V_{y_n}\}$ of Y.

Then $U = \bigcap_{i=1}^{n} U_{y_i}$ intersects none of the V_{y_i}, and so does not intersect Y.

Example: non-closed intervals in $\mathbb{R}$ are not compact.

We will show (later) all $[a,b]$ $(a, b \in \mathbb{R})$ are compact.

Remark: If $Y \subseteq X$ is compact and X is Hausdorff, and $x_0 \notin Y$, $\exists$ open sets U, V s.t. $x_0 \in U$, $Y \subseteq V$.